The Visual Computer (2025) 41:12679–12690
https://doi.org/10.1007/s00371-025-04180-5

RESEARCH

SAMirror: enhancing mirror detection via integrated visual-depth cues in segment anything model

Qiushi Meng[1] · Yunxun Liu[1] · Ran Hu[1] · Mengmeng Liang[1] · Jixue Yan[1] · Lei Zhu[2]

Received: 11 June 2025 / Accepted: 1 September 2025 / Published online: 16 September 2025

Abstract

Mirror detection remains a significant challenge in computer vision due to misleading reflections and the lack of distinctive visual cues. While the segment anything model (SAM) demonstrates strong segmentation capabilities, its performance on mirrors is limited by its reliance on RGB features alone. We introduce SAMirror, a novel framework that integrates rich visual information from SAM with crucial depth cues for robust mirror detection. Our core innovation lies in the depth-infusion adapter, which injects and contextualizes auxiliary scene context. Extensive experiments on mirror detection datasets demonstrate that SAMirror outperforms state-of-the-art methods, achieving an IoU of 0.902 and an F-measure of 0.930 on the MSD dataset, showcasing superior performance in both accuracy and generalization. By effectively leveraging depth information, SAMirror advances the field of mirror detection, offering new possibilities for applications in robotics, autonomous navigation, and beyond..

Keywords Mirror detection · Segment anything model · Mixture of experts

1 Introduction

Detecting mirrors is a long-standing and crucial problem in computer vision due to its wide applicability in real-world environments. For example, in robotics and embodied AI, the inability to distinguish between traversable spaces and reflective surfaces can lead to catastrophic navigation failures, such as a robot mistaking a mirror for a clear path. Moreover, in tasks such as 3D reconstruction and SLAM, unrecognized mirrors can distort spatial understanding by introducing phantom geometry or duplicate structures. Therefore, robust mirror detection is essential for safe and intelligent perception in diverse applications ranging from autonomous navigation and human–robot interaction to indoor scene understanding.

For the mirror detection task, numerous methods [3, 4, 17, 22, 28, 34, 39] have been proposed over the years, ranging from handcrafted features to deep learning-based models. Early approaches often relied on geometric cues, such as specular highlights [3] to infer the presence of mirrors, but they tend to fail under complex lighting or cluttered backgrounds. Recent learning-based methods [4, 17, 28, 34, 39] significantly improve performance by leveraging semantic context and boundary information, yet they still struggle in highly reflective scenes where appearance cues are misleading or ambiguous.

The advent of large-scale foundation models such as segment anything model (SAM) [11] opens new opportunities for tackling this task, achieving strong generalization and zero-shot capabilities across diverse domains. Among them, segment anything model (SAM) [11] and its enhanced version SAM2 [25] have demonstrated remarkable performance by leveraging massive-scale image-text data and powerful

✉ Lei Zhu
leizhu@ust.hk

Qiushi Meng
mqs19941018@gmail.com

Yunxun Liu
1049353387@qq.com

Ran Hu
hur@guangzhou.csg.cn

Mengmeng Liang
Liang341204@qq.com

Jixue Yan
yan_jix@126.com

1 Guangzhou Power Supply Bureau of Guangdong Power Grid Co., Ltd, Guangzhou, China

2 HKUST, Hong Kong, China & HKUST(GZ), Guangzhou, China

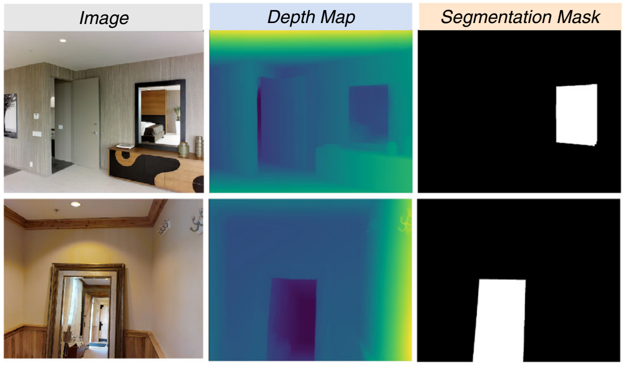

Fig. 1 As shown in the depth map (middle), mirror regions frequently exhibit depth discontinuities at their boundaries, as the reflected scene's depth does not correspond to the actual physical environment. Our SAMirror leverages this insight to enhance mirror detection capability by effectively infusing depth priors into a powerful visual backbone

vision transformers. However, SAM is designed to segment based on visual prompts and high-level semantics, and may still fall short in scenarios where appearance alone is insufficient for accurate segmentation, such as mirrors, which visually mimic their surrounding content [9].

To compensate for this, we investigate whether incorporating geometric cues such as depth can provide strong complementary information. Unlike RGB appearance, depth is less affected by reflectance and can reveal discontinuities or physical inconsistencies at mirror boundaries. This motivates us to explore how to effectively infuse depth priors into powerful vision foundation models like SAM to enhance their mirror detection capability. As shown in Fig. 1, mirror regions often produce depth discontinuities at their boundaries, since the reflected scene does not correspond to the actual 3D structure of the environment. These discontinuities offer a reliable geometric hint for detecting mirrors. Leveraging this characteristic provides new possibilities for enhancing mirror segmentation models with depth-aware understanding.

Therefore, in this paper, we propose SAMirror, a novel framework that leverages both powerful semantic priors from foundation models and complementary geometric cues from depth information. Our core idea is to adapt and enhance SAM to perform accurate mirror segmentation by integrating depth signals in a principled and efficient manner.

Specifically, we design a depth-infusion adapter, which seamlessly injects depth cues into the frozen visual encoder of SAM with minimal computational overhead. This adapter modulates intermediate visual features using spatially aligned depth representations, enabling the model to distinguish reflective surfaces from actual scene content. To further improve the model's capacity for complex contextual reasoning, we introduce a perceptual mixture of experts unit, which dynamically activates specialized expert branches based on perceptual patterns, thus enhancing multi-scale contextual understanding crucial for resolving appearance ambiguity. Moreover, we develop a contextual consensus decoder, which hierarchically fuses the enriched features and progressively refines the mirror segmentation mask. This decoder leverages a multi-scale supervision strategy to ensure that both coarse semantic and fine-grained boundary cues are preserved throughout the decoding process. Extensive experiments on multiple public mirror segmentation benchmarks validate the effectiveness of our framework. SAMirror consistently outperforms state-of-the-art methods by a significant margin, demonstrating the value of integrating geometric priors into vision foundation models for challenging perception tasks.

Our key contributions are summarized as follows:

- We propose the first framework to fuse SAM with depth information for robust mirror surface detection.
- We propose a depth-infusion adapter, which strategically injects crucial depth cues into the visual backbone with minimal computational overhead.
- We introduce a perceptual mixture of experts unit, designed to profoundly enrich multi-scale contextual understanding via an adaptive expert gating mechanism.
- We design a contextual consensus decoder, which hierarchically aggregates and refines the enriched features to progressively reconstruct the precise mirror segmentation mask, facilitated by robust multi-scale supervision.
- Extensive experiments on public mirror segmentation benchmarks demonstrate that SAMirror outperforms state-of-the-art methods

2 Related work

2.1 Mirror detection

Mirror detection aims to accurately identify and delineate mirror regions from a single natural image, which is crucial for real-world applications such as autonomous driving and robotic perception. Accurate mirror detection can effectively prevent accidents caused by specular reflections.

In recent years, semantic segmentation methods have seen steady progress, evolving from early fully convolutional networks [20] to advanced models like DeepLabV3 [1], which employs atrous convolutions and ASPP to effectively capture multi-scale context. Mask2Former [2] further advances this by combining transformer architectures with mask prediction, improving the modeling of complex semantics. However, these methods often underperform in mirror detection. Due to the highly reflective nature of mirrors, they typically present the reflected external environment rather than possessing intrinsic visual features. This results in strong semantic and textural similarity between mirror and non-mirror regions, rendering traditional saliency-based and edge-based cues less effective. In addition, mirrors often lack distinctive boundaries, color contrast, or shape patterns, further complicating their accurate identification.

To address these challenges, Yang et al. [39] proposed MirrorNet, the first model specifically designed for mirror detection. They also introduced a dedicated mirror detection dataset and a context-contrast mechanism between mirror interior and exterior regions. However, MirrorNet's reliance on contextual inconsistency becomes a limitation in cases where the reflected and surrounding scenes are highly similar. To overcome this, Lin et al. [17] proposed PMDNet, incorporating a multi-scale edge enhancement module that emphasizes structural cues at mirror boundaries and established a more challenging benchmark for the task, advancing the field. Guan et al. [4] utilized graph convolutional networks (GCNs) to model contextual dependencies among image regions, enhancing discriminability between mirrors and background. Building further, Tan et al. [28] introduced VCNet, which explicitly modeled geometric correspondences across mirror boundaries by leveraging visual chirality cues and structural symmetry, achieving robust performance in complex scenes. EAPT [18] employs deformable attention to preserve spatial context across scales, which has proven effective in segmentation tasks.. Despite recent advances, these methods highly rely on supervised learning and finely labeled datasets, which limits their performance in complex real-world scenarios.

2.2 Segment Anything Model

Segment anything model (SAM) [11] represents a major milestone in the field of vision foundation models for segmentation, which can segment any object in the image based on interaction prompts. SAM has shown impressive generalizability in a wide range of image segmentation tasks [8, 14, 30, 35, 36]. Based on SAM, EfficientSAM [37] adopts its masked image pre-training strategy to build a lightweight ViT encoder, significantly reducing computational complexity while retaining segmentation performance comparable to SAM. In addition, MedSAM [21] improves SAM's medical image segmentation capability by fine-tuning it on a large-scale dataset. Segment Anything Model 2 (SAM2) [25] extends SAM by adopting a transformer-based architecture augmented with a stream-based memory mechanism. It also introduces a hierarchical image encoder, which enhances feature representation while maintaining compatibility with lightweight task-specific decoders.

However, recent studies have shown that both SAM and SAM2 perform poorly in mirror detection. Han et al. [6] systematically evaluated the performance of SAM in mirror and transparent object segmentation, revealing significant limitations due to the high semantic and textural similarity between the mirror and surrounding regions. Similarly, SAM2 has shown unstable performance in video-based mirror detection tasks, especially when relying solely on sparse prompts [9]. These challenges underscore the need for a task-specific adaptation strategy to enhance mirror detection performance in complex scenes.

2.3 Depth Estimation

Depth estimation techniques have significantly advanced the understanding of scene structure, which is especially beneficial in challenging cases involving reflection, occlusion, or complex geometry. Li et al. [12] propose RADepthNet, a reflectance-aware monocular depth estimation framework that explicitly models surface reflectivity, thus enhancing depth prediction accuracy in scenes with reflective materials like mirrors. In industrial applications, TAMDepth [13] introduces a hybrid self-supervised monocular depth estimation model that captures both local details and global context by integrating multi-scale convolutional modules with transformer blocks. DepthAnything-V2 [38] leverages image-scale training for generalized monocular depth estimation.

Moreover, some works underscore the importance of explicitly modeling depth and scene structure to improve perceptual tasks [40]. Kim et al. [10] demonstrate a real-time reconstruction pipeline for curved pipes using RGBD cameras, leveraging depth data to accurately infer 3D shapes from superpixel decompositions and parametric modeling. Sheng et al. [27] presents a depth-aware motion deblurring method based on loopy belief propagation, illustrating how depth priors can constrain and improve image restoration in challenging visual scenarios. Inspired by these insights, our work extends this paradigm to mirror detection: We embed depth discontinuity signals within a powerful foundation segmentation model (SAM) to overcome RGB ambiguity and improve the precision of reflective surface segmentation.

3 Method

3.1 Overview

The overall architecture of our proposed SAMirror is depicted in Fig. 2. The network takes an input image $I \in \mathbb{R}^{3\times H\times W}$ and produces a pixel-level mirror segmentation map $S \in \mathbb{R}^{H\times W}$. SAMirror incorporates three key components: a depth-infusion adapter that injects auxiliary depth cues into the visual backbone in a lightweight yet effective manner, enhancing depth awareness throughout the feature hierarchy, a perceptual mixture of experts (PMoE) unit that dynamically aggregates multi-scale contextual information via an adaptive expert gating mechanism,

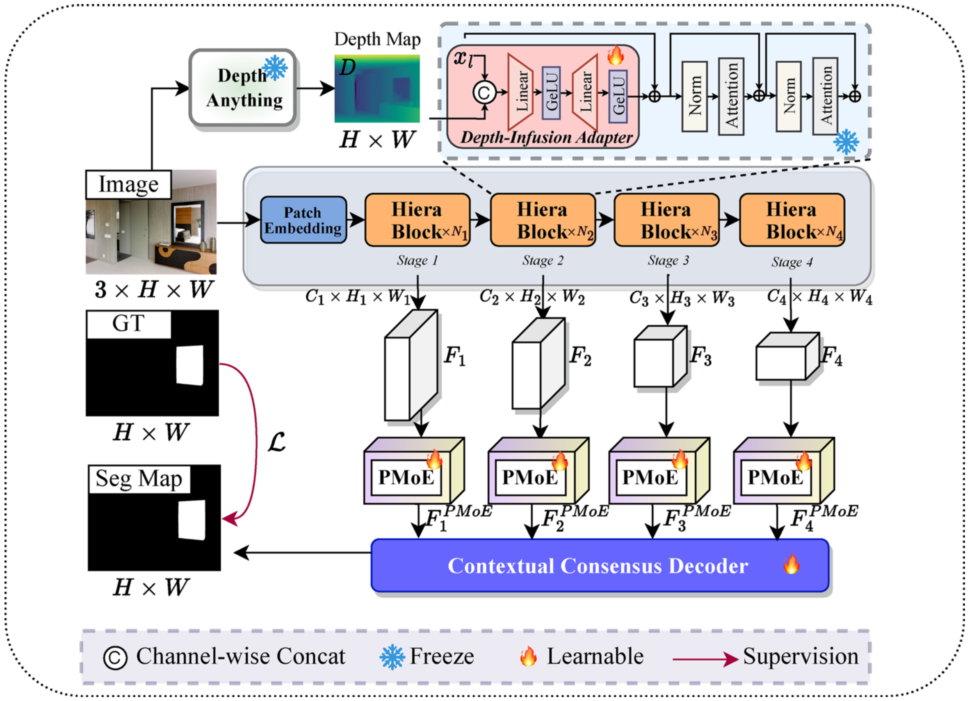

Fig. 2 Overview of the proposed SAMirror framework. Given an RGB image and its corresponding depth map, our model first extracts visual features using the SAM2 backbone. The proposed depth-infusion adapter strategically injects depth cues into intermediate stages of SAM2 to enhance feature representation for reflective surfaces. Then, the perceptual mixture of experts unit adaptively fuses multi-scale features to handle the complex context and ambiguous mirror boundaries. Finally, the contextual consensus decoder hierarchically aggregates and refines the features to produce accurate mirror segmentation masks

Table 1 Comparison with other methods on three benchmark datasets. "↑" indicates higher is better, while "↓" means lower is better

Dataset	Models	Publication	IoU ↑	ACC ↑	F_β ↑	MAE ↓	BER ↓
MSD [39]	MirrorNet [39]	ICCV'2019	0.845	0.948	0.892	0.044	7.67
	PMDNet [17]	CVPR'2020	0.847	0.952	0.908	0.033	7.22
	Segformer [31]	NIPS'2021	0.879	0.953	0.915	0.038	6.50
	SANet [4]	CVPR'2022	0.871	0.951	0.912	0.033	7.10
	VCNet [28]	TPAMI'2022	0.884	0.958	0.914	0.029	7.32
	HetNet [7]	AAAI'2023	0.888	0.964	0.922	0.027	6.63
	CSFwinformer [32]	TIP'2024	0.875	0.960	0.918	0.030	6.79
	Ours	-	**0.902**	**0.968**	**0.930**	**0.023**	**5.92**
PMD [17]	MirrorNet [39]	ICCV'2019	0.648	0.798	0.785	0.030	10.39
	PMDNet [17]	CVPR'2020	0.658	0.818	0.795	0.028	10.86
	Segformer [31]	NIPS'2021	0.678	0.833	0.808	0.024	10.14
	SANet [4]	CVPR'2022	0.657	0.816	0.794	0.024	10.97
	VCNet [28]	TPAMI'2022	0.676	0.818	0.801	0.025	10.16
	HetNet [7]	AAAI'2023	0.702	0.827	0.818	0.023	9.65
	CSFwinformer [32]	TIP'2024	0.705	0.820	0.815	0.029	9.93
	Ours	-	**0.729**	**0.839**	**0.829**	**0.020**	**8.82**
RGBD- Mirror [23]	Mask2Former [2]	ICCV'2019	0.768	0.861	0.803	0.040	12.16
	PMDNet [17]	CVPR'2020	0.739	0.819	0.774	0.035	13.24
	Segformer [31]	NIPS'2021	0.754	0.858	0.788	0.039	13.11
	SANet [4]	CVPR'2022	0.749	0.841	0.789	0.039	12.88
	VCNet [28]	TPAMI'2022	0.763	0.844	0.810	0.034	12.28
	HetNet [7]	AAAI'2023	0.775	0.857	0.819	0.031	11.20
	CSFwinformer [32]	TIP'2024	0.769	0.850	0.815	0.034	11.04
	Ours	-	**0.792**	**0.867**	**0.836**	**0.026**	**10.02**

and a contextual consensus decoder that hierarchically fuses and refines the enriched features from different scales, progressively decoding a high-resolution, boundary-preserving mirror segmentation mask with the assistance of multi-scale supervision.

3.2 Depth-infusion adapter

Reflective surfaces, such as mirrors, pose a unique challenge to conventional image segmentation models due to their ambiguous appearance and context-mimicking nature. Unlike ordinary objects, mirrors lack consistent color, texture, or edge cues, often blending seamlessly into their surroundings. However, depth cues provide a crucial complementary signal: mirrors typically exhibit anomalous or absent depth responses due to their reflective properties. Despite this, off-the-shelf segmentation frameworks, especially foundation models like SAM [25], are agnostic to such geometric cues, limiting their effectiveness in mirror perception.

3.2.1 Encoder

Initially, we freeze all parameters of SAM2's Hiera encoder. Then, we use SAM2's Hiera encoder to extract multi-scale semantic features. The encoder produces a feature pyramid $F_L \in \mathbb{R}^{B\times H_L\times W_L\times C_L}(L = \{1, 2, 3, 4\})$, where deeper levels capture increasingly abstract representations at lower resolutions ($H_L = \frac{H}{2^{i+1}}$, $W_L = \frac{W}{2^{i+1}}$).

3.2.2 Depth prior generation

To obtain reliable depth estimations for in-the-wild images, we employ Depth Anything v2 [38] as an off-the-shelf monocular depth predictor. Given the input image $I \in \mathbb{R}^{B\times 3\times H\times W}$, the corresponding dense depth map $D \in \mathbb{R}^{B\times 1\times H\times W}$ is inferred and used as complementary input throughout the encoder pathway.

For the F_L and its corresponding depth map, $D \in \mathbb{R}^{B\times 1\times H\times W}$. The depth map D is first spatially interpolated via bilinear sampling to match the resolution of F_L, yielding $D_L \in \mathbb{R}^{B\times 1\times H_L\times W_L}$. D_L is then reshaped and concatenated with X_L along the channel dimension, forming $F_L^{concat} \in \mathbb{R}^{B\times H_L\times W_L\times (C_L+1)}$. This concatenated tensor is then fed into a sequence of linear layers with GeLU activations. This network generates a depth-conditioned adaptation vector as $\Delta F_L \in \mathbb{R}^{B\times H_L\times W_L\times C_L}$.

3.2.3 Feature augmentation

Finally, the adaptation vector ΔF_L is added element-wise to the original SAM2 feature F_L, producing the depth-infused feature

$$F_L' = F_L + \Delta F_L. \tag{1}$$

This adaptive mechanism allows SAMirror to align SAM2's powerful visual features with task-specific depth priors, effectively transforming a general segmenter into a specialized, depth-aware feature extractor explicitly tailored for discerning reflective surfaces. By freezing the original SAM2 encoder weights and only training the depth-infusion adapter, we maintain the strong generalized representations while minimizing additional training overhead.

3.3 Perceptual mixture of experts unit

Mirror detection, encompassing objects of highly varying scales and contexts, necessitates a robust mechanism for multi-scale feature aggregation. To address this, we introduce the perceptual mixture of experts unit (PMoE). Unlike conventional cascaded decoders prone to information loss during successive propagations, our PMoE explicitly leverages a dynamic expert routing mechanism to adaptively emphasize the most relevant multi-scale features for each input. This approach is inspired by the success of mixture of experts [26] designs in large-scale models, which allow for efficient model scaling and specialized feature learning.

As illustrated in Fig. 3, for an input feature map $F_L' \in \mathbb{R}^{B\times C_L\times H_L\times W_L}$, the PMoE employs four parallel expert branches. Each branch $\mathcal{E}_j(\cdot)$ $(j = 1, 2, 3, 4)$ is meticulously designed to extract features with distinct receptive field characteristics through varying dilation rates (e.g., 1, 3, 5, 7) using dilated convolutions. This yields a set of expert-specific feature representations $\{E_0, E_1, E_2, E_3\}$, each focusing on different scales of contextual information.

The core of the PMoE lies in its adaptive gating mechanism. A lightweight gating network $\mathcal{G}(\cdot)$ processes the input feature F_L' to dynamically generate a set of scalar weights $W \in \mathbb{R}^{B\times 4}$. This network typically comprises a global pooling layer followed by compact convolutional layers and a softmax activation, ensuring that the weights for each input sum to one. These weights W_j represent the importance of each expert branch for a given input region.

The output is obtained by a weighted sum of the expert features, as

$$F_L'' = \sum_{j=1}^{4} W_j \cdot F_j, \tag{2}$$

where $\cdot$ denotes pixel-wise multiplication, enabling a dynamic combination of the expert outputs. A residual connection is further applied for enhanced stability as

$$F_L^{PMoE} = F_L'' + F_L'. \tag{3}$$

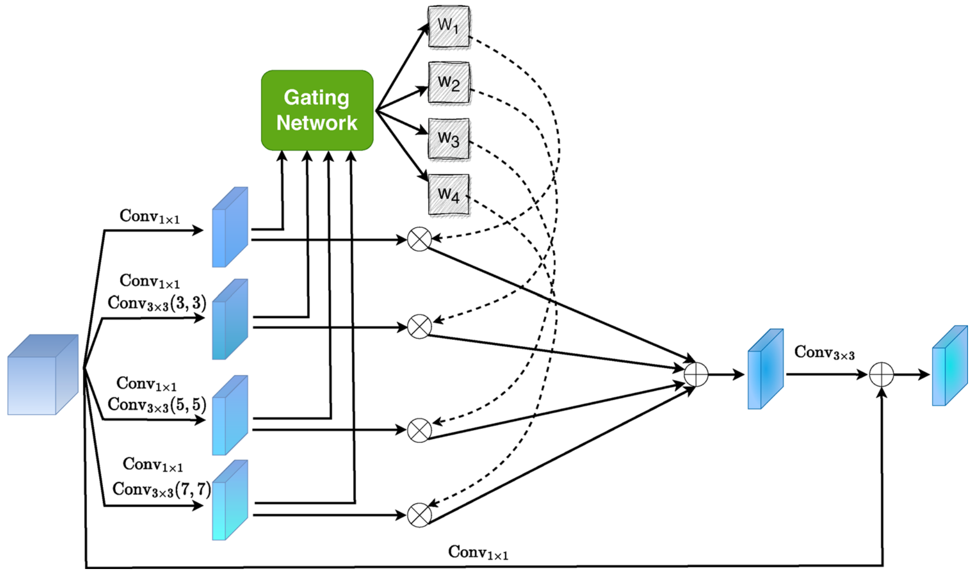

Fig. 3 Illustration of the perceptual mixture of experts (PMoE) unit. Given multi-scale features, PMoE uses parallel dilated convolution branches and a gating network to adaptively fuse features from different receptive fields. Then, a learned gating mechanism adaptively selects and fuses expert pathways specialized for different spatial and contextual patterns

Table 2 Comparison with other methods on video mirror segmentation dataset (VMD-D [15]). "↑" indicates higher is better, while "↓" means lower is better

Models	IoU ↑	ACC ↑	F_β ↑	MAE ↓
PMDNet [17]	0.532	0.872	0.749	0.128
VCNet [28]	0.539	0.877	0.749	0.123
HetNet [7]	0.531	0.868	0.748	0.131
VMD-Net [15]	0.567	0.895	0.787	0.105
Ours	**0.592**	**0.908**	**0.812**	**0.092**

This adaptive fusion empowers SAMirror to prioritize and synthesize features from the most pertinent receptive fields, thereby improving the network's ability to precisely delineate mirror boundaries across a spectrum of sizes and complexities.

3.4 Contextual Consensus Decoder

Although the segmentation backbone provides multi-scale feature representations, directly merging these features often leads to coarse boundaries and inconsistent activations, especially for reflective surfaces where appearance varies drastically across scales. To address this, we propose the contextual consensus decoder (CCD), which hierarchically aggregates multi-scale features while establishing cross-scale contextual alignment through attention-based refinement.

Specifically, the decoder reconstructs the final prediction through a sequence of feature fusion and upsampling steps, designed to progressively recover spatial details and semantically refine the segmentation boundaries. For each decoding stage $i \in \{3, 2, 1\}$, the operation integrates the upsampled higher-level features with lateral encoder features via concatenation followed by convolutional refinement, as

$$\tilde{F}_{L+1} = \mathrm{Up}\left(F_{L+1}^{PMoE}\right), \tag{4}$$

$$\tilde{F}'_L = \phi_i\left(\mathrm{Concat}\left(\tilde{F}_{L+1}, F_L^{PMoE}\right)\right), \tag{5}$$

where $\tilde{F}_{L+1} \in \mathbb{R}^{C_{i+1}\times\frac{H}{2^{i+1}}\times\frac{W}{2^{i+1}}}$ is the higher-level decoder output, $\mathrm{Up}(\cdot)$ denotes bilinear upsampling by a factor of 2, and $\phi_i(\cdot)$ is a convolutional block, followed by batch normalization and ReLU activation, that refines the fused feature map. This fusion process yields the intermediate decoder features $\{\tilde{F}'_1, \tilde{F}'_2, \tilde{F}'_3\}$, with increasing spatial resolution and decreasing semantic abstraction. At each stage, a prediction head $\mathcal{H}_i(\cdot)$ is applied to produce an auxiliary segmentation map, as

$$\hat{S}_i = \mathcal{H}_i\left(\tilde{F}'_L\right), \quad i \in \{2, 3\}, \tag{6}$$

The head $\mathcal{H}_i(\cdot)$ consists of a 1×1 convolution followed a sigmoid function. Each $\hat{S}_i$ is then upsampled to the original input resolution for deep supervision. The final segmentation map $\hat{S}$ is produced from the last decoded feature $\mathcal{F}_1$ through a dedicated head:

$$\hat{S} = \mathcal{H}_1\left(\tilde{F}'_1\right). \tag{7}$$

This hierarchical decoding structure enables the network to reach a consensus across multiple levels of contextual abstraction, supporting both global structure awareness and boundary-level precision, which is essential for accurate mirror segmentation.

3.5 Multi-scale supervision

The network is trained with hierarchical supervision at all three output levels. The loss function is summarized as:

$$\mathcal{L} = \lambda_1 \cdot \mathcal{L}_{\text{seg}}(\hat{S}_1, \mathcal{G}) + \lambda_2 \cdot \mathcal{L}_{\text{seg}}(\hat{S}_2, \mathcal{G}) + \lambda_3 \cdot \mathcal{L}_{\text{seg}}(\hat{S}, \mathcal{G}), \tag{8}$$

where $\mathcal{G}$ is the ground-truth mirror mask, and λ_1, λ_2, λ_3 are loss weights empirically set to balance the gradients. Following the approaches in [29], we use the weighted IoU loss and BCE loss as our segmentation supervised training objectives, i.e., $\mathcal{L}_{\text{seg}}$. This design ensures that high-resolution details from early layers are preserved and aligned with deep semantic features, leading to accurate and coherent segmentation of reflective surfaces.

4 Experiment

4.1 Datasets and benchmarks

We adopt three commonly used annotated datasets, MSD [39],[1] PMD [17],[2] and RGBD-Mirror [23],[3] to construct the training set following [33]. The combined training set can alleviate overfitting caused by limited data in individual datasets. For fair comparison, we retrain all comparison methods using their official code on the combined training data, tuning parameters to achieve the best performance.

4.2 Experimental setup

4.2.1 Implementation details

We use pre-trained DepthAnything-V2 Small [38] to predict the depth estimation maps, and we utilize the pre-trained SAM2-Hiera-tiny [25] pre-trained on SA-1B as the encoder. We implement our framework using PyTorch and conduct all experiments on a computing server equipped with four NVIDIA 4090 GPUs. The training process spans 15 epochs, employing the Adam optimizer with a learning rate of 3×10^{-4} and a weight decay of 1×10^{-4}. Each input image is resized to a fixed resolution of 512×512, and the batch size is set to 12. The loss function weight $\lambda_1, \lambda_2, and \lambda_3$ is set to 0.25, 0.25, and 0.5.

[1] https://mhaiyang.github.io/ICCV2019_MirrorNet/index

[2] https://jiaying.link/cvpr2020-pgd/

[3] https://mhaiyang.github.io/CVPR2021_PDNet/index

Table 3 Computational cost comparison with SOTA methods in terms of trainable parameters and inference speed (FPS) on 512 × 512 inputs using a NVIDIA 4090 GPU

Models	FPS ↑	Params (M) ↓
VCNet [28]	18.67	333.43
PMDNet [17]	10.80	147.65
HetNet [7]	27.68	49.87
SAMirror (full)	**53.48**	**46.77**

4.2.2 Evaluation metrics

Following [5, 16, 19, 24, 28], we utilize intersection over union (IoU), pixel accuracy (Accuracy), F-measure (F_β), mean absolute error (MAE), and balanced error rate (BER) as the evaluation metrics. The IoU is adopted to evaluate the degree of overlap between ground truths and prediction maps as

$$IoU = \frac{TP}{TP + FP + FN}, \tag{9}$$

where the TN, TP, and FN denote the true negative, true positive, and false negative pixels, respectively. Accuracy measures the overall correctness of pixel-wise classification in segmentation tasks, as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{10}$$

F_β is a average of Precision and Recall, as

$$F_\beta = \frac{(1 + \beta^2) Precision \times Recall}{\beta^2 Precision + Recall}, \tag{11}$$

where $Precision = \frac{TP}{TP+FP}$ and $Recall = \frac{TP}{TP+FN}$.

MAE is the absolute error of the predictions and ground truths, as

$$\text{MAE} = \frac{1}{H \times W} \sum_{x=1}^{W} \sum_{y=1}^{H} |P(x, y) - G(x, y)|, \tag{12}$$

Balance error rate (BER) is the average of the false positive and false negative rates, as

$$BER = 1 - 0.5 \times (\frac{N_{TP}}{N_P} + \frac{N_{TN}}{N_N}), \tag{13}$$

where N_{TP}, N_P, N_{TN}, N_N are the number of true positives, true negatives, object and non-object pixels, respectively.

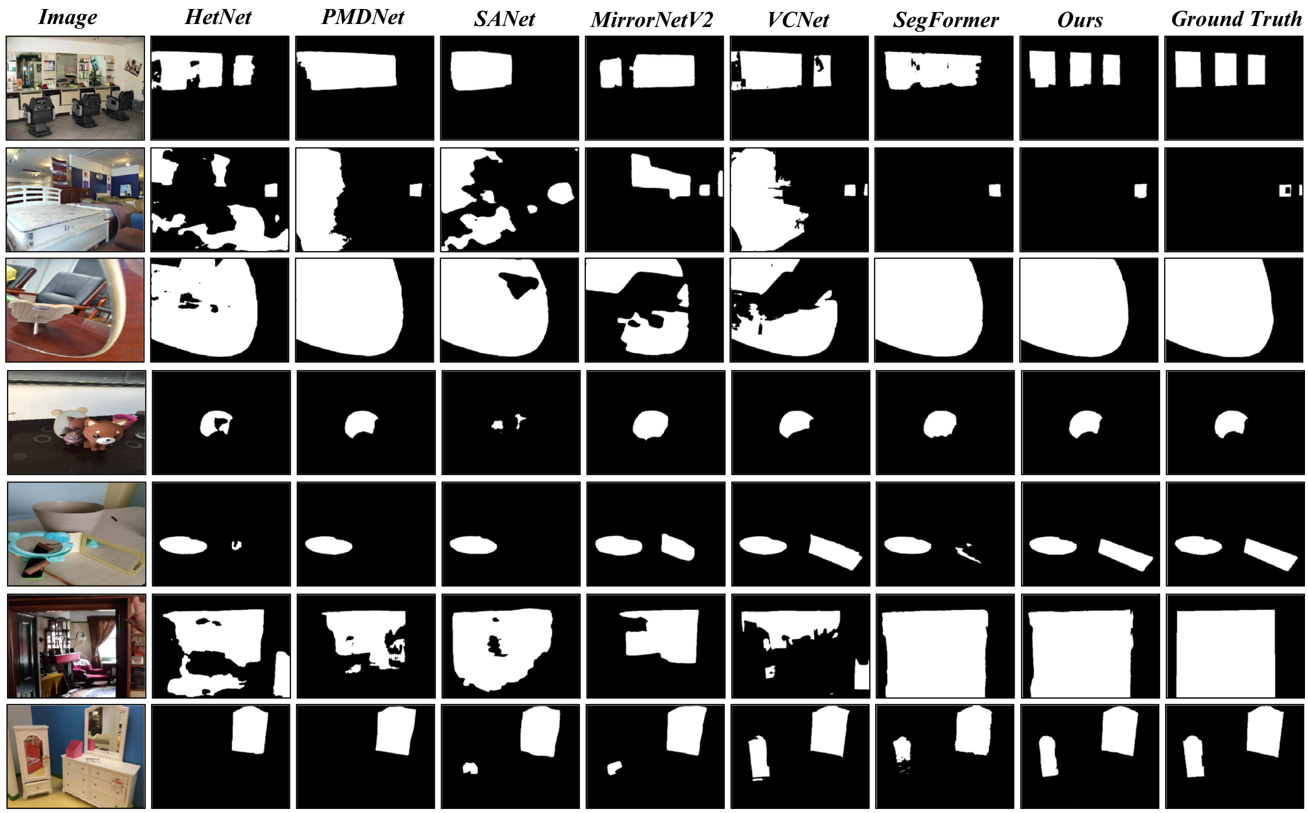

Fig. 4 Qualitative comparisons between our proposed SAMirror and state-of-the-art mirror detection methods

4.3 Comparisons with state-of-the-art methods

4.3.1 Compared methods

We compare our method with seven state-of-the-art networks, including commonly used CNN-based models such as MirrorNet [39], PMDNet [17], SANet [4], VCNet [28], and HetNet [7], as well as recent transformer-based methods such as SegFormer [31] and CSFwinformer [32].

4.3.2 Quantitative results

Table 1 presents the quantitative comparison between our proposed SAMirror and other mirror detection methods. Across all datasets and metrics, our method consistently achieves the best performance, demonstrating the effectiveness of our unified framework. Specifically, on the MSD dataset, our model achieves an IoU of 0.902, F_β of 0.930, and a MAE of only 0.023, outperforming the closest competitor HetNet [7] by a notable margin. This confirms our model's strong capability in capturing fine boundaries and challenging transparent regions in complex scenes. On the PMD dataset, which emphasizes progressive mirror segmentation, our method attains an IoU of 0.729 and the lowest BER of 8.82, indicating robust performance under weak or incomplete mirror cues. Similarly, on the RGBD-Mirror dataset, where the depth modality plays a crucial role, our approach surpasses all baselines with an F_β score of 0.836 and MAE of 0.026, verifying the benefit of depth-aware segmentation.

To assess the generalization ability of our approach beyond static images, we further evaluate it on a benchmark video mirror segmentation dataset, VMD-D [15]. As shown in Table 2, our method outperforms previous state-of-the-art approaches on the video mirror segmentation dataset, achieving higher IoU, ACC, and F_β, and lower MAE. This demonstrates that our approach generalizes well to dynamic scenes despite being trained primarily on static images.

4.3.3 Qualitative results

To further evaluate the effectiveness of our proposed framework, we demonstrate the visual comparisons between our SAMirror and other compared methods in Fig. 4. Our method consistently produces more accurate and coherent segmentation masks across various challenging scenes.

Specifically, in cluttered environments with strong specular reflections or complex mirror surfaces, previous methods often suffer from over-segmentation or miss fine-grained boundary details. In contrast, our framework leverages the semantic awareness of the SAM2 backbone and the cross-

Table 4 Ablation study of different components on the MSD [39] dataset. Each module incrementally improves the performance, demonstrating its effectiveness

Model Variant	IoU ↑	Accuracy ↑	F_β ↑	MAE ↓	BER ↓
+ Contextual Consensus Decoder (CCD)	0.858	0.952	0.903	0.036	7.21
+ Depth-Infusion Adapter	0.874	0.958	0.912	0.031	6.70
+ Perceptual Mixture of Experts Unit (PMoE)	0.883	0.961	0.918	0.029	6.40
+ CCD + PMoE	0.886	0.962	0.920	0.028	6.33
SAMirror	**0.894**	**0.964**	**0.926**	**0.027**	**6.12**

Table 5 Number of trainable parameters for each component

Model Variant	Trainable Params (M)
CCD	0.71
Depth-Infusion Adapter	0.87
PMoE	0.88
SAMirror (full)	1.96

modal guidance from depth priors to effectively distinguish mirrors from other regions. For instance, in the sixth row of Fig. 4, most comparison methods incorrectly classify the mirror boundary, while our model precisely captures the subtle boundaries and produces an accurate mask.

4.4 Ablation studies

To validate the effectiveness of each key component in our proposed SAMirror, we conduct ablation studies on the MSD [39] dataset. As shown in Table 4, we progressively introduce each module to evaluate its individual and cumulative impact on performance.

We begin with a baseline model that leverages the SAM2 backbone with the contextual consensus decoder (CCD) but without incorporating depth cues. The results demonstrate the effectiveness of hierarchical refinement and multi-scale supervision. Adding the depth-infusion adapter significantly boosts performance, highlighting the importance of injecting depth information to disambiguate mirrors from their surroundings. Incorporating the perceptual mixture of experts (PMoE) leads to further improvement by dynamically aggregating multi-scale semantic context. The complete integration of all proposed components achieves the highest performance, confirming that each module offers distinct and complementary contributions to the final outcome. To further examine the effectiveness of parameter-efficient fine-tuning strategies, we additionally evaluate a variant where the SAM2 encoder is entirely frozen, and no adapter modules or other learnable parameters are introduced inside the encoder. In this configuration, only the CCD and PMoE are trainable. As reported in Table 4, this variant ("+ CCD + PMoE") achieves an IoU of 0.886 and an F_β score of 0.920 on the MSD dataset. The comparison with the + CCD" baseline (IoU = 0.858, F_β = 0.903) confirms that the proposed PMoE module brings consistent gains even under a basic parameter-efficient fine-tuning setting.

Computational Efficiency We compare SAMirror with representative SOTA methods in terms of inference speed (FPS) and model size. The results in Table 3 show that SAMirror achieves the highest FPS (53.48) among all methods, while also maintaining a compact parameter size (46.77 M). This demonstrates that our design not only improves segmentation accuracy, but also delivers real-time performance, which is crucial for latency-sensitive applications such as robotics and autonomous navigation.

Number of Learnable Parameters. To further analyze the efficiency of our framework, we report the number of trainable parameters for each key component in Table 5. The results show that SAMirror achieves substantial performance gains with only 1.85M trainable parameters in total, highlighting its parameter-efficient design. Specifically, the contextual consensus decoder (CCD), depth-infusion adapter, and perceptual mixture of experts unit (PMoE) introduce 0.33M, 0.26M, and 1.25M parameters, respectively. This compact parameter budget enables SAMirror to deliver state-of-the-art accuracy while maintaining low computational overhead, making it well-suited for real-time and resource-constrained applications.

5 Conclusion

In this work, we presented SAMirror, a novel mirror segmentation framework that effectively bridges RGB and depth cues to address the inherent challenges of reflective surface segmentation within the powerful SAM2 architecture. First, we introduce a depth-infusion adapter, which strategically embeds depth awareness into the visual backbone with minimal overhead. Second, we propose a perceptual mixture of experts unit to enhance multi-scale contextual representation via adaptive expert selection conditioned on the input semantics. Third, we develop a contextual consensus decoder, which hierarchically aggregates these enriched features to achieve accurate and coherent mirror segmentation. Extensive experiments on benchmark datasets validate the effectiveness of our approach, demonstrating significant improvements over existing state-of-the-art methods.SAMirror can also be integrated into SLAM pipelines by masking detected mirror regions in RGBD frames before depth fusion, thereby preventing phantom geometry and improving mapping accuracy.

Acknowledgements This project was supported by the China Southern Power Grid Science and Technology Project (Project No.: 030117KC 23120005).

Author Contributions Qiushi Meng, Yunxun Liu, Ran Hu, Mengmeng Liang, and Jixue Yan have their contributions in writing the code, running the experiments, and drafting the paper.
Lei Zhu is responsible for supervising the whole paper, and revising the paper.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Chen, L.C., Papandreou, G., Kokkinos, I., Murphy, K., Yuille, A.L.: Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. IEEE Trans. Pattern Anal. Mach. Intell. **40**(4), 834–848 (2017)
2. Cheng, B., Choudhuri, A., Misra, I., Kirillov, A., Girdhar, R., Schwing, A.G.: Mask2former for video instance segmentation. arXiv preprint (2021) arXiv:2112.10764
3. DelPozo, A., Savarese, S.: Detecting specular surfaces on natural images. In: 2007 IEEE Conference on Computer Vision and Pattern Recognition, pp. 1–8. IEEE (2007)
4. Guan, H., Lin, J., Lau, R.W.: Learning semantic associations for mirror detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5941–5950 (2022)
5. Guo, D., Li, K., Hu, B., Zhang, Y., Wang, M.: Benchmarking microaction recognition: dataset, method, and application. IEEE Trans. Circuits Syst. Video Technol. **34**(7), 6238–6252 (2024)
6. Han, D., Zhang, C., Qiao, Y., Qamar, M., Jung, Y., Lee, S., Bae, S.H., Hong, C.S.: Segment anything model (sam) meets glass: Mirror and transparent objects cannot be easily detected. arXiv preprint (2023) arXiv:2305.00278
7. He, R., Lin, J., Lau, R.W.: Efficient mirror detection via multi-level heterogeneous learning. In: Proceedings of the AAAI Conference on Artificial Intelligence, **37**, pp. 790–798 (2023)
8. Huang, D., Xiong, X., Ma, J., Li, J., Jie, Z., Ma, L., Li, G.: Alignsam: Aligning segment anything model to open context via reinforcement learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3205–3215 (2024)
9. Jie, L.: When sam2 meets video shadow and mirror detection. arXiv preprint (2024) arXiv:2412.19293
10. Kim, D.M., Ahn, J., Kim, S., Lee, J., Kim, M., Han, J.: Real-time reconstruction of pipes using rgb-d cameras. Comput. Animat. Virtual Worlds **35**(1), e2197 (2024)
11. Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., et al.: Segment anything. In: Proceedings of the IEEE/CVF international conference on computer vision, pp. 4015–4026 (2023)
12. Li, C., Yi, R., Ali, S.G., Ma, L., Wu, E., Wang, J., Mao, L., Sheng, B.: Radepthnet: Reflectance-aware monocular depth estimation. Virtual Reality & Intell. Hardware **4**(5), 418–431 (2022)
13. Li, S., Lyu, C., Xia, B., Chen, Z., Zhang, L.: Tamdepth: self-supervised monocular depth estimation with transformer and adapter modulation. Vis. Comput. **40**(10), 6797–6808 (2024)
14. Li, W., Xiong, X., Xia, P., Ju, L., Ge, Z.: Tp-drseg: improving diabetic retinopathy lesion segmentation with explicit text-prompts assisted sam. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 743–753. Springer (2024)
15. Lin, J., Tan, X., Lau, R.W.: Learning to detect mirrors from videos via dual correspondences. In: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 9109–9118 (2023)
16. Lin, J., Tan, X., Lau, R.W.: Learning to detect mirrors from videos via dual correspondences. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9109–9118 (2023)
17. Lin, J., Wang, G., Lau, R.W.: Progressive mirror detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3697–3705 (2020)
18. Lin, X., Sun, S., Huang, W., Sheng, B., Li, P., Feng, D.D.: Eapt: efficient attention pyramid transformer for image processing. IEEE Trans. Multimedia **25**, 50–61 (2021)
19. Liu, L., Prost, J., Zhu, L., Papadakis, N., Liò, P., Schönlieb, C.B., Aviles-Rivero, A.I.: Scotch and soda: A transformer video shadow detection framework. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10449–10458 (2023)
20. Long, J., Shelhamer, E., Darrell, T.: Fully convolutional networks for semantic segmentation. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 3431–3440 (2015)
21. Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B.: Segment anything in medical images. Nat. Commun. **15**, 654 (2024)
22. Madeira, T., Oliveira, M., Dias, P.: Reflection-aware 3d mirror segmentation and pose estimation. The Visual Computer pp. 1–10 (2024)
23. Mei, H., Dong, B., Dong, W., Peers, P., Yang, X., Zhang, Q., Wei, X.: Depth-aware mirror segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3044–3053 (2021)
24. Mei, H., Liu, Y., Wei, Z., Zhou, D., Wei, X., Zhang, Q., Yang, X.: Exploring dense context for salient object detection. IEEE Trans. Circuits Syst. Video Technol. **32**(3), 1378–1389 (2021)
25. Ravi, N., Gabeur, V., Hu, Y.T., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., et al.: Sam 2: Segment anything in images and videos. arXiv preprint (2024) arXiv:2408.00714
26. Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J.: Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In: International Conference on Learning Representations (2017)
27. Sheng, B., Li, P., Fang, X., Tan, P., Wu, E.: Depth-aware motion deblurring using loopy belief propagation. IEEE Trans. Circuits Syst. Video Technol. **30**(4), 955–969 (2019)
28. Tan, X., Lin, J., Xu, K., Chen, P., Ma, L., Lau, R.W.: Mirror detection with the visual chirality cue. IEEE Trans. Pattern Anal. Mach. Intell. **45**(3), 3492–3504 (2022)
29. Wei, J., Wang, S., Huang, Q.: F^3net: fusion, feedback and focus for salient object detection. In: AAAI, pp. 12321–12328 (2020)
30. Wu, J., Ji, W., Liu, Y., Fu, H., Xu, M., Xu, Y., Jin, Y.: Medical sam adapter: Adapting segment anything model for medical image segmentation (2023)
31. Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J.M., Luo, P.: Segformer: Simple and efficient design for semantic segmentation

with transformers. Adv. Neural. Inf. Process. Syst. **34**, 12077–12090 (2021)
32. Xie, Z., Wang, S., Yu, Q., Tan, X., Xie, Y.: Csfwinformer: Cross-space-frequency window transformer for mirror detection. IEEE Transactions on Image Processing (2024)
33. Xing, Z., Liu, L., Yang, Y., Wang, H., Ye, T., Chen, S., Li, W., Liu, G., Zhu, L.: Detect any mirrors: Boosting learning reliability on large-scale unlabeled data with an iterative data engine. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2025)
34. Xing, Z., Liu, L., Ye, T., Chen, S., Yang, Y., Liu, G., Xu, X., Zhu, L.: Farther than mirror: Explore pattern-compensated depth of mirror with temporal changes for video mirror detection
35. Xiong, X., Wang, C., Li, W., Li, G.: Mammo-sam: Adapting foundation segment anything model for automatic breast mass segmentation in whole mammograms. In: International Workshop on Machine Learning in Medical Imaging, pp. 176–185. Springer (2023)
36. Xiong, X., Wu, Z., Tan, S., Li, W., Tang, F., Chen, Y., Li, S., Ma, J., Li, G.: Sam2-unet: Segment anything 2 makes strong encoder for natural and medical image segmentation. arXiv preprint (2024) arXiv:2408.08870
37. Xiong, Y., Varadarajan, B., Wu, L., Xiang, X., Xiao, F., Zhu, C., Dai, X., Wang, D., Sun, F., Iandola, F., et al.: Efficientsam: Leveraged masked image pretraining for efficient segment anything. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 16111–16121 (2024)
38. Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., Zhao, H.: Depth anything v2. In: Advances in Neural Information Processing Systems, vol. 37, pp. 21875–21911. Curran Associates, Inc. (2024)
39. Yang, X., Mei, H., Xu, K., Wei, X., Yin, B., Lau, R.W.: Where is my mirror? In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 8809–8818 (2019)
40. Zhu, X., Yao, X., Zhang, J., Zhu, M., You, L., Yang, X., Zhang, J., Zhao, H., Zeng, D.: Tmsdnet: Transformer with multi-scale dense network for single and multi-view 3d reconstruction. Comput. Animat. Virtual Worlds **35**(1), e2201 (2024)

Qiushi Meng received his M.S. degree in Electrical Engineering from North China Electric Power University in 2021. He currently serves as the Digital Transmission Technology Specialist at Guangzhou Power Supply Bureau of Guangdong Power Grid Co., Ltd., China Southern Power Grid. His research interest includes digital transmission technologies and computer vision applications.

Yunxun Liu graduated from North China Electric Power University (Baoding) in 2017 and then worked at Guangzhou Power Supply Bureau of Guangdong Power Grid Co., Ltd., China Southern Power Grid. He mainly focused on transmission line operation and maintenance, intelligent transmission construction, and technology innovation-related businesses.

Ran Hu received his M.S. degree in Electrical Engineering from China Three Gorges University in 2015. He is currently working at Guangzhou Power Supply Bureau of Guangdong Power Grid Co., Ltd., China Southern Power Grid, where he is engaged in production technology management for the operation and maintenance of transmission lines. His professional expertise primarily includes transmission line operation and maintenance technology, as well as the application of digitalization and artificial intelligence in power grid operations.

Mengmeng Liang received his Master's degree in Electrical Engineering from the School of Electrical and Electronic Engineering, Huazhong University of Science and Technology in 2017. His research interest included the operation and maintenance management of power transmission equipment, as well as research and application of power transmission technology.

Jixue Yan received her M.S. degree in Electrical Engineering from South China University of Technology in 2015. She currently serves as the Production Planning Specialist at Guangzhou Power Supply Bureau of Guangdong Power Grid Co., Ltd., China Southern Power Grid. Her research interest includes the operation and maintenance management of transmission equipment, research and application of transmission technology, and digital application of transmission.

Lei Zhu received his Ph.D. degree in the Department of Computer Science and Engineering from the Chinese University of Hong Kong in 2017. He is currently working as an assistant professor at the ROAS Thrust & the DSA Thrust, The Hong Kong University of Science and Technology (Guangzhou), and also an affiliated assistant professor at the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology. Before that, he was a postdoctoral researcher at DAMTP, University of Cambridge. His research interests include computer vision, image restoration, image enhancement, image and video processing, medical AI, and deep learning.